Unit - 02 :

$V_1 \cdot V_2 = 0$
$V_1 \cdot V_3 = 0$ } orthogonal $V_i \cdot V_j = 0$ $i \neq j$
$V_2 \cdot V_3 = 0$ orthogonal

$V_1 \cdot V_1 = 1$
$V_2 \cdot V_2 = 1$ } $i = j$ orthogonal
$V_3 \cdot V_3 = \phi$

verify which of the following vectors are orthogonal

① $S_1 = \{ \underbrace{(-1, 4, -3)}_{v_1}, \underbrace{(5, 2, 1)}_{v_2}, \underbrace{(3, -4, -7)}_{v_3} \}$

$V_1 \cdot V_2 = -5 + 8 - 3 \Rightarrow 0$
$V_2 \cdot V_3 = 15 - 8 - 7 \Rightarrow 1 \neq 0 \rightarrow S_1$ is not orthogonal set
$V_1 \cdot V_3 = -3 - 16 + 21 \Rightarrow 2 \neq 0$ } not required

② $S_2 = \{ (2, -3), (6, 4) \}$
$V_1 \cdot V_2 \Rightarrow 12 - 12 \Rightarrow 0$
$\therefore S_2$ is orthogonal vector

③ $\begin{bmatrix} \frac{-3}{\sqrt{11}} & -1/\sqrt{6} & -1/\sqrt{66} \\ 1/\sqrt{11} & 2/\sqrt{6} & \frac{-4}{\sqrt{66}} \\ 1/\sqrt{11} & 1/\sqrt{6} & -7/\sqrt{66} \end{bmatrix}$ verify orthogonal are not.

→ Consider here each column considered as vector

$V_1 \cdot V_2 = \frac{-3}{\sqrt{6}\sqrt{11}} + \frac{2}{\sqrt{11}\sqrt{6}} + \frac{1}{\sqrt{11 \times 6}} \Rightarrow 0$

~~$\therefore$ [ ] is not a orthogonal.~~

$V_1 \cdot V_3 \Rightarrow \frac{-3}{\sqrt{11} \cdot \sqrt{66}} - \frac{4}{\sqrt{11} \cdot \sqrt{66}} - \frac{7}{\sqrt{66} \cdot \sqrt{11}} \neq 0$

$\therefore$ not a orthogonal vector

(c) Verify $\begin{bmatrix} 3/\sqrt{11} & -1/\sqrt{6} & -1/\sqrt{66} \\ 1/\sqrt{11} & 2/\sqrt{6} & -4/\sqrt{66} \\ 1/\sqrt{11} & 1/\sqrt{6} & 7/\sqrt{66} \end{bmatrix}$

Verify orthogonal or orthonormal or not

$$v_1 \cdot v_2 = \frac{-3}{\sqrt{66}} + \frac{2}{\sqrt{66}} + \frac{1}{\sqrt{66}} \Rightarrow 0$$

$$v_1 \cdot v_3 = \frac{-3}{\sqrt{11\times 66}} - \frac{4}{\sqrt{11\times 66}} + \frac{7}{\sqrt{11\times 66}} \Rightarrow 0$$

$$v_2 \cdot v_3 = \frac{1}{\sqrt{6\times 66}} - \frac{8}{\sqrt{6\times 66}} + \frac{7}{\sqrt{6\times 66}} \Rightarrow 0$$

$\therefore$ It is orthogonal

$$v_1 \cdot v_1 = \left(\frac{3}{\sqrt{11}}\right)^2 + \left(\frac{1}{\sqrt{11}}\right)^2 + \left(\frac{1}{\sqrt{11}}\right)^2 \Rightarrow \frac{9}{11} + \frac{1}{11} + \frac{1}{11}$$

$$\Rightarrow \frac{11}{11} \Rightarrow 1$$

$$v_2 \cdot v_2 = \frac{1}{6} + \frac{4}{6} + \frac{1}{6} \Rightarrow \frac{6}{6} \Rightarrow 1$$

$$v_3 \cdot v_3 = \frac{1}{66} + \frac{16}{66} + \frac{49}{66} \Rightarrow \frac{66}{66} \Rightarrow 1$$

$\therefore v_1 \cdot v_1 = v_2 \cdot v_2 = v_3 \cdot v_3 = 1$

$\therefore$ It is a orthonormal vector

Constructing orthogonal Basis set:

Given Set $S = \{x_1, x_2, x_3, x_4\}$

$$v_1 = x_1$$

$$v_2 = x_2 - \left[\frac{x_2 \cdot v_1}{v_1 \cdot v_1}\right] v_1$$

$$v_3 = x_3 - \left[\frac{x_3 v_1}{v_1 \cdot v_1}\right] v_1 - \left(\frac{v_3 \cdot v_2}{v_2 \cdot v_2}\right) v_2$$

(1) Construct an orthogonal basis ~~for~~ $\{v_1, v_2\}$ for Sol $\{x_1, x_2\}$ where $x_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$ $x_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$

Sol $x_1 = (3, 6, 0)$ $x_2 (1, 2, 2)$

$v_1 = x_1 \Rightarrow (3, 6, 0)$

$v_2 = x_2 - \left[\frac{x_2 \cdot v_1}{v_1 \cdot v_1}\right] v_1$

$= (1, 2, 2) - \left[\frac{(3\cdot1)+(6\cdot2)+(0\cdot2)}{(3\cdot3)+(6\cdot6)+(0\cdot0)}\right] v_1$

$= (1, 2, 2) - \left[\frac{15}{9+36}\right] \Rightarrow \frac{15}{45} \cdot v_1$

$= (1, 2, 2) - \frac{\cancel{15}^{1}}{\cancel{45}_{3}} (3, 6, 0)$

$= (1, 2, 2) - (1, 2, 0)$

$v_2 = (0, 0, 2)$

hence $\{v_1, v_2\} = \left\{ \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} \right\}$

$v_1 \cdot v_2 = 0$

$= 3\cdot0 + 6\cdot0 + 0\cdot2 \Rightarrow 0$

$\therefore$ v_1 & v_2 are orthogonal.

for ~~$v_1 \cdot v_1 =$~~ orthonormal:

$(3, 6, 0) = \left(\frac{3}{\sqrt{3^2+6^2+0^2}}, \frac{6}{\sqrt{3^2+6^2+0^2}}, \frac{0}{\sqrt{3^2+6^2+0^2}}\right)$

$= \left(\frac{3}{\sqrt{45}}, \frac{6}{\sqrt{45}}, \frac{0}{\sqrt{45}}\right)$

$(0, 0, 2) \longrightarrow \cancel{(0, 0, 1)} = \sqrt{0+0+4} = 2$

$= \left(0, 0, \frac{2}{2}\right) \Rightarrow (0, 0, 1)$

$\therefore \left\{ \begin{bmatrix} 3/\sqrt{45} \\ 6/\sqrt{45} \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

(d) Cons orthonormal basis for Columns of $\boxed{A}$

$$A = \begin{array}{c} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \\ \begin{matrix} x_1 & x_2 & x_3 \end{matrix} \end{array}$$

$v_1 = x_1 \Rightarrow (1, 1, 1, 1)$

$$v_2 = x_2 - \left[\frac{x_2 v_1}{v_1 \cdot v_1}\right] v_1$$

$$= (0, 1, 1, 1) - \left[\frac{0+1+1+1}{1+1+1+1}\right] (1, 1, 1, 1)$$

$$= (0, 1, 1, 1) - \left[\frac{3}{4}\right] (1, 1, 1, 1)$$

$$v_2 = \left(-\frac{3}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$$

$1 - \frac{3}{4}$

$$v_3 = x_3 - \left[\frac{x_3 v_1}{v_1 \cdot v_1}\right] v_1 - \left[\frac{x_3 \cdot v_2}{v_2 \cdot v_2}\right] v_2$$

$\frac{\cancel{2}\,1}{\cancel{4}\,2}$

$$= (0, 0, 1, 1) - \left[\frac{0+0+1+1}{4}\right] v_1 - \left[\frac{0+0+\frac{1}{4}+\frac{1}{4}}{\frac{9}{16}+\frac{1}{16}+\frac{1}{16}+\frac{1}{16}}\right] v_2$$

$$= (0, 0, 1, 1) - \left[\frac{1}{2}\right] (1, 1, 1, 1) - \left[\frac{1/2}{\frac{\cancel{12}\,6\,3}{\cancel{16}\,8\,4}}\right] \left(-\frac{3}{4}, \frac{1}{4}, \ldots\right)$$

$$= (0, 0, 1, 1) - \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right) -$$

$\frac{1}{\cancel{2}} \cdot \frac{\cancel{4}^{2}}{3} \Rightarrow \frac{2}{3} v_2$

$$\left(\frac{\cancel{2}^{1}}{\cancel{3}} \times \frac{-\cancel{3}}{\cancel{4}_{2}}, \frac{\cancel{2}}{3} \times \frac{1}{\cancel{4}_{2}}, \frac{\cancel{2}}{3} \times \frac{1}{\cancel{4}_{2}}, \frac{\cancel{2}}{3} \times \frac{1}{\cancel{4}_{2}}\right)$$

$-\frac{1}{2} - \frac{1}{6}$

$\frac{-3-1}{6} = \frac{-\cancel{4}^{2}}{\cancel{6}_{3}}$

$$= (0, 0, 1, 1) - \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right) - \left(-\frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}\right)$$

$$= \left(-\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right) - \left(-\frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}\right)$$

$$v_3 = \left(0, -\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

Side calculations:

$\frac{1}{2} - \frac{1}{6}$

$\frac{3-1}{6} \Rightarrow \frac{\cancel{2}\,1}{\cancel{6}\,3}$

$-\frac{1}{2} - \frac{1}{6} = \frac{-3-1}{6}$

$-\frac{\cancel{4}\,2}{\cancel{6}\,3}$

Orthogonal set $\{v_1, v_2, v_3\} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}, \begin{bmatrix} 0 \\ -2/3 \\ 1/3 \\ 1/3 \end{bmatrix} \right\}$

$v_1 \cdot v_2 = -\frac{3}{4} + \frac{3}{4} = 0$

$v_2 \cdot v_3 = -\frac{2}{12} + \frac{2}{12} \Rightarrow 0$

$v_1 \cdot v_3 \Rightarrow -\frac{2}{3} + \frac{2}{3} \Rightarrow 0$

$\therefore$ v_1, v_2, v_3 are orthogonal set.

<u>for orthonormal</u>

$\sqrt{1^2+1^2+1^2+1^2} \Rightarrow \sqrt{4} \Rightarrow 2 \quad = \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}$

$\sqrt{\left(-\frac{3}{4}\right)^2 + \left(\frac{1}{4}\right)^2 + \left(\frac{1}{4}\right)^2 + \left(\frac{1}{4}\right)^2}$

$= \sqrt{\cancel{\frac{12}{16}}\frac{3}{4}} \Rightarrow \frac{\sqrt{3}}{2} \Rightarrow \begin{bmatrix} -\frac{3}{\cancel{4}_2} \times \frac{\cancel{2}^1}{\sqrt{3}} = \frac{-3}{2\sqrt{3}} \\ \frac{1}{\cancel{4}_2} \times \frac{\cancel{2}}{\sqrt{3}} \\ \frac{1}{2\sqrt{3}} \\ \frac{1}{2\sqrt{3}} \end{bmatrix} = \begin{bmatrix} -\frac{3}{2\sqrt{3}} \\ \frac{1}{2\sqrt{3}} \\ \frac{1}{2\sqrt{3}} \\ \frac{1}{2\sqrt{3}} \end{bmatrix}$

$-\frac{\sqrt{3} \times \sqrt{3}}{2\sqrt{3}} = -\frac{\sqrt{3}}{2}$

$= \sqrt{\left(-\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2} = \sqrt{\frac{6}{9}}$

$= \frac{\sqrt{6}}{3} \Rightarrow \begin{bmatrix} 0 \\ -\frac{2}{\cancel{3}} \times \frac{\cancel{3}}{\sqrt{6}} \\ \frac{1}{3} \times \frac{3}{\sqrt{6}} \\ \frac{1}{3} \times \frac{3}{\sqrt{6}} \end{bmatrix} \Rightarrow \begin{bmatrix} 0 \\ -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}$

(Q) Find an orthogonal basis

for Col $[\ ]^{**} = \begin{bmatrix} -1 & 6 & 6 \\ 3 & -8 & 3 \\ 1 & -2 & 6 \\ 1 & -4 & -3 \end{bmatrix}$

$\quad\quad\quad x_1 \quad x_2 \quad x_3$

$V_1 = x_1 \Rightarrow (-1, 3, 1, 1)$

$$V_2 = x_2 - \left[\frac{x_2 V_1}{V_1 \cdot V_1}\right] V_1$$

$$= (6, -8, -2, -4) - \left[\frac{-6 - 24 - 2 - 4}{1 + 9 + 1 + 1}\right] V_1$$

$$= (6, -8, -2, -4) - \left[-\frac{\cancel{36}\,3}{\cancel{12}\,1}\right](-1, 3, 1, 1)$$

$$= (6, -8, -2, -4) - (3, -9, -3, -3)$$

$$V_2 = (3, 1, 1, -1)$$

$$V_3 = x_3 - \left[\frac{x_3 \cdot V_1}{V_1 \cdot V_1}\right] V_1 - \left[\frac{x_3 \cdot V_2}{V_2 \cdot V_2}\right] V_2$$

$$= (6, 3, 6, -3) - \left[\frac{\cancel{-6 + 9 + 6 - 3}}{\cancel{12}\,2}\right] V_1 - \left[\frac{18 + 3 + 6 + 3}{9 + 1 + 1 + 1}\right]$$

$$\left[\frac{1}{2}\right] V_1 - \frac{\cancel{30}\,\cancel{10}\,5}{\cancel{12}\,\cancel{4}\,2} V_2 = \frac{5}{2} V_2$$

$$\boxed{= (6, 3, 6, -3) - \left(-\frac{1}{2}, \frac{3}{2}, \frac{1}{2}, \frac{1}{2}\right) - \left(\frac{15}{2}, \frac{5}{2}, \frac{5}{2}, -\frac{5}{2}\right)}$$

$$= (6, 3, 6 - 3) - \left(\frac{\cancel{16}\,7}{\cancel{2}}, \frac{\cancel{8}\,4}{\cancel{2}}, \frac{\cancel{6}\,3}{\cancel{2}}, \frac{\cancel{4}\,2}{\cancel{2}}\right)$$

$$= (6, 3, 6, -3) + (-7, -4, -3, 2)$$

$-\frac{1}{2} - \frac{1}{2}$

$\frac{3}{2} - \frac{5}{2}$

$$V_3 = (14, 4, 8, -6)$$

$$\{v_1, v_2, v_3\} = \left\{ \begin{bmatrix} -1 \\ 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 3 \\ -1 \end{bmatrix} \right\}$$

$V_1 \cdot V_2 = -3+3+1-1 = 0$

$V_1 \cdot V_3 \Rightarrow 1-3+3-1 \Rightarrow 0$

$V_2 \cdot V_3 \Rightarrow -3-1+3+1 \Rightarrow \underline{0}$

$\therefore V_1, V_2$ & V_3 are orthogonal set

for orthonormal

$\sqrt{1^2+3^2+1^2+1} \Rightarrow \sqrt{12} \Rightarrow \sqrt{4\times 3} \Rightarrow 2\sqrt{3}$

$\sqrt{9+1+1+1} \Rightarrow \sqrt{12} \to 2\sqrt{3}$

$\sqrt{1+1+9+1} = \sqrt{12} \to 2\sqrt{3}$

$$= \left\{ \begin{bmatrix} -1/2\sqrt{3} \\ 3/2\sqrt{3} \\ 1/2\sqrt{3} \\ 1/2\sqrt{3} \end{bmatrix} \begin{bmatrix} 3/2\sqrt{3} \\ 1/2\sqrt{3} \\ 1/2\sqrt{3} \\ -1/2\sqrt{3} \end{bmatrix} \begin{bmatrix} -1/2\sqrt{3} \\ -1/2\sqrt{3} \\ 3/2\sqrt{3} \\ -1/2\sqrt{3} \end{bmatrix} \right\}$$

$\therefore A = Q \cdot R \longrightarrow$ upper Triangular matrix

orthogonal matrix

$A = QR$

$Q^T A = R$

$\boxed{R = Q^T A}$

$A_{n\times n}$ orthogonal $\to A^T A = I$

$A = QR$ and Q is orthogonal.

① Find Q-R Factorization of $A = \begin{array}{c} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \\ x_1 \quad x_2 \quad x_3 \end{array}$

By Gram-Schmidth orthogonal we have soln Previous

Ans we solve from above

for Orthonormal, we have soln

$$Q = \begin{bmatrix} 1/2 & -\frac{3}{2\sqrt{3}} & 0 \\ 1/2 & 1/2\sqrt{3} & -2/\sqrt{6} \\ 1/2 & 1/2\sqrt{3} & 1/\sqrt{6} \\ 1/2 & 1/2\sqrt{3} & 1/\sqrt{6} \end{bmatrix}$$

$V_1 = x_1 \Rightarrow (-1, 3, 1, 1)$

$V_2 = x_2 - \left[\frac{x_2 V_1}{V_1 \cdot V_1}\right] V_1$

$= (6, -8, -2, -4) - \left[\frac{-6 - 24 - 2 - 4}{1 + 9 + 1 + 1}\right] V_1$

$= (6, -8, -2, -4) - \left[\frac{-36}{12}\right] [-1, 3, 1, 1]$

$= (6, -8, -2, -4) - (3, -9, -3, -3)$

$V_2 = (3, 1, 1, -1)$

$V_3 = x_3 - \left[\frac{x_3 \cdot V_1}{V_1 \cdot V_1}\right] V_1 - \left[\frac{x_3 \cdot V_2}{V_2 \cdot V_2}\right] V_2$

$= (6, 3, 6, -3) - \left[\frac{-6 + 9 + 6 - 3}{12}\right] V_1 - \left[\frac{18 + 3 + 6 + 3}{9 + 1 + 1 + 1}\right]$

$\left[\frac{1}{2}\right] V_1 - \frac{30}{12} V_2 = \frac{5}{2} V_2$

$\boxed{= (6, 3, 6, -3) - \left(-\frac{1}{2}, \frac{3}{2}, \frac{1}{2}, \frac{1}{2}\right) - \left(\frac{15}{2}, \frac{5}{2}, \frac{5}{2}, -\frac{5}{2}\right)}$

$= (6, 3, 6 - 3) - \left(\frac{16}{2}, \frac{8}{2}, \frac{6}{2}, \frac{4}{2}\right)$

$= (6, 3, 6, -3) + (-7, -4, -3, 2)$

$V_3 = (14, 4, 8, -6)$

$\{V_1, V_2, V_3\} = \left\{ \begin{bmatrix} -1 \\ 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 3 \\ -1 \end{bmatrix} \right\}$

$V_1 \cdot V_3 \Rightarrow 1 - 3 + 3 - 1 \Rightarrow 0$

$V_2 \cdot V_3 \Rightarrow -3 - 1 + 3 + 1 \Rightarrow 0$

$\therefore V_1, V_2$ & V_3 are orthogonal set

for orthonormal

$\sqrt{1^2 + 3^2 + 1^2 + 1} \Rightarrow \sqrt{12} = \sqrt{4 \times 3} \Rightarrow 2\sqrt{3}$

$\sqrt{9 + 1 + 1 + 1} \Rightarrow \sqrt{12} \Rightarrow 2\sqrt{3}$

$\sqrt{1 + 1 + 9 + 1} = \sqrt{12} \Rightarrow 2\sqrt{3}$

$= \left\{ \begin{bmatrix} -1/2\sqrt{3} \\ 3/2\sqrt{3} \\ 1/2\sqrt{3} \\ 1/2\sqrt{3} \end{bmatrix} \begin{bmatrix} 3/2\sqrt{3} \\ 1/2\sqrt{3} \\ 1/2\sqrt{3} \\ -1/2\sqrt{3} \end{bmatrix} \begin{bmatrix} -1/2\sqrt{3} \\ -1/2\sqrt{3} \\ 3/2\sqrt{3} \\ -1/2\sqrt{3} \end{bmatrix} \right\}$

$\therefore A = QR \longrightarrow$ upper Triangular matrix

orthogonal matrix

$A = QR$

$Q^T A = R$

$\boxed{R = Q^T A}$

$A_{n \times n}$ orthogonal $\rightarrow A^T A = I$

$A = QR$ and Q is orthogonal.

① Find QR Factorization of $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ (x_1, x_2, x_3)

By Gramschmidth orthogonal we have solv Previous Ans we solve from doubt

For orthonormal, we have soln $Q = \begin{bmatrix} 1/2 & -\frac{3}{2\sqrt{3}} & 0 \\ 1/2 & 1/2\sqrt{3} & -2/\sqrt{6} \\ 1/2 & 1/2\sqrt{3} & 1/\sqrt{6} \\ 1/2 & 1/2\sqrt{3} & 1/\sqrt{6} \end{bmatrix}$

Now $A = QR$

Operate with Q^T

$Q^T.A = R$

$Q^T.A = \underbrace{Q^T.Q}_{I}\, R$

$\underline{R = Q^T.A}$

$$Q^T.A = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2\sqrt{3}} & \frac{1}{2\sqrt{3}} & \frac{1}{2\sqrt{3}} \\ 0 & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$-\frac{3}{2\sqrt{3}}$

$$= \begin{bmatrix} 2 & \frac{3}{2} & 1 \\ 0 & \frac{3}{2\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & 0 & \frac{2}{\sqrt{6}} \end{bmatrix}$$

(2) Find Q-R factorization of A.

$$A = \begin{array}{c} \begin{bmatrix} 1 & 2 & 5 \\ -1 & 1 & -4 \\ -1 & 4 & -3 \\ 1 & -4 & 7 \\ 1 & 2 & 1 \end{bmatrix} \\ \begin{matrix} x_1 \quad & x_2 \quad & x_3 \end{matrix} \end{array}$$

(1) $v_1 = x_1 = (1, -1, -1, 1, 1)$

$v_2 = x_2 - \left(\frac{x_2 \cdot v_1}{v_1 \cdot v_1}\right) \cdot v_1 \qquad \left(\frac{-5}{5} \Rightarrow -1\right)$

$= (2, 1, 4, -4, 2) - \left(\frac{\cancel{2} - 1 - 4 - \cancel{4} + \cancel{2}}{5}\right)(1, -1, -1, 1, 1)$

$v_2 = (3, 0, 3, -3, 3)$

① obtain Q-R factorization of

$$A = \begin{array}{c} \begin{bmatrix} 1 & -1 & 4 \\ 1 & 4 & -2 \\ 1 & 4 & 2 \\ 1 & -1 & 0 \end{bmatrix} \\ \begin{matrix} x_1 & x_2 & x_3 \end{matrix} \end{array}$$

Let $x_1 = (1,1,1,1)$

$x_2 = (-1,4,4,-1)$

$x_3 = (4,-2,2,0)$

$v_1 = a_1 = (1,1,1,1)$

$$v_2 = x_2 - \left[\frac{x_2 \cdot v_1}{v_1 \cdot v_1}\right] \cdot v_1 \rightarrow \frac{6}{4} = \frac{3}{2}$$

$$= (-1,4,4,-1) - \left[\frac{-1+4+4-1}{4}\right](1,1,1,1)$$

$$= (-1,4,4,-1) - \left(\frac{3}{2}, \frac{3}{2}, \frac{3}{2}, \frac{3}{2}\right)$$

$$v_2 = \left(-\frac{5}{2}, \frac{5}{2}, \frac{5}{2}, -\frac{5}{2}\right) \quad \checkmark \ v_2$$

$$v_3 = x_3 - \left(\frac{x_3 \cdot v_1}{v_1 \cdot v_1}\right) v_1 - \left(\frac{x_3 \cdot v_2}{v_2 \cdot v_2}\right) v_2 \rightarrow \left(\frac{2}{5}\right)\left(-\frac{5}{2}, \frac{5}{2}, \frac{5}{2}, -\frac{5}{2}\right)$$

$$v_3 = (4,-2,2,0) - \left(\frac{4-2+2+0}{4}\right)(v_1) - \left(\frac{4\cdot\frac{5}{2} + -2\cdot\frac{5}{2} + 2\cdot\frac{5}{2} + 0\cdot\frac{5}{2}}{\frac{100}{4}}\right)$$

$$\Downarrow 1 \qquad \frac{2}{5} = \frac{10}{25} \qquad \frac{100}{4} = 25$$

$$v_3 = (4,-2,2,0) - (1,1,1,1) + \frac{2}{5}\left(-\frac{5}{2}, \frac{5}{2}, \frac{5}{2}, -\frac{5}{2}\right)$$

$$v_3 = (4,-2,2,0) + \left(\frac{3}{2}, -\frac{7}{2}, -\frac{7}{2}, \frac{3}{2}\right) \qquad -1+\frac{5}{2} \quad -1-\frac{5}{2}$$

$$v_3 = (3,-3,1,-1) + \frac{2}{5}\left(-\frac{5}{2}, \frac{5}{2}, \frac{5}{2}, -\frac{5}{2}\right)$$

$$v_3 = (2,-2,2,-2)$$

$v_1 \cdot v_2 = 0, \quad v_2 \cdot v_3 = 0, \quad v_1 \cdot v_3 = 0$

$$u_1 = \frac{v_1}{|v_1|} = \frac{(1,1,1,1)}{\sqrt{1^2+1^2+1^2+1^2}} \Rightarrow 2 \Rightarrow \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$$

$$u_2 = \frac{v_2}{|v_2|} = \frac{-\frac{5}{2}, \frac{5}{2}, \frac{5}{2}, -\frac{5}{2}}{\sqrt{\frac{100}{4}}} \quad 25 \Rightarrow 5 \quad = \left(-\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right)$$

$$u_3 = \left(\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right) = \frac{(2,-2,2,-2)}{\sqrt{16}} \Rightarrow 4$$

$$Q = [u_1 \; u_2 \; u_3] = \begin{bmatrix} 1/2 & -1/2 & 1/2 \\ 1/2 & 1/2 & -1/2 \\ 1/2 & 1/2 & 1/2 \\ 1/2 & -1/2 & -1/2 \end{bmatrix}$$

$$R = Q^T . A$$

$$= \begin{bmatrix} 1/2 & 1/2 & 1/2 & 1/2 \\ -1/2 & 1/2 & 1/2 & -1/2 \\ 1/2 & -1/2 & 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} 1 & -1 & 4 \\ 1 & 4 & -2 \\ 1 & 4 & 2 \\ 1 & -1 & 0 \end{bmatrix}$$

$$R = \begin{bmatrix} 2 & 3 & 2 \\ 0 & 5 & -2 \\ 0 & 0 & 4 \end{bmatrix}$$

others Notes

$Ax = b$ where $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix}$

$$(A^T . A)x = A^T . b$$

$~~LHS~~$ $5x + 3y = 0$

$(-)\,3x + 3y = 6$

$2x = -6$

$\boxed{y = 5}$ $\boxed{x = -3}$

ii) minimum distance

$Ax = b$

Approximate soln $\hat{x}$

$$A\hat{x} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} -3 \\ 5 \end{bmatrix}$$

$$d = \sqrt{1^2 + 2^2 + 1^2}$$

$$d = \sqrt{6}$$

$$A . \hat{x} = \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}$$

$$d = (5, 2, -1) - \overset{b}{(6, 0, 0)}$$

$$d = (-1, 2, -1)$$

1) Obtain a 6. Soln by least approximation method the $Ax=b$ where $A=\begin{bmatrix}1&0&1\\1&1&-1\\1&2&-3\end{bmatrix}$, $b=\begin{bmatrix}6\\0\\0\end{bmatrix}$

$$(A^T \cdot A) \cdot x = A^T b$$

$$\begin{bmatrix}1&1&1\\0&1&2\\1&-1&-3\end{bmatrix}\begin{bmatrix}1&0&1\\1&1&-1\\1&2&-3\end{bmatrix} x = \begin{bmatrix}1&1&1\\0&1&2\\1&-1&-3\end{bmatrix}\begin{bmatrix}6\\0\\0\end{bmatrix}$$

$$\begin{bmatrix}3&3&-3\\3&5&-7\\-3&-7&11\end{bmatrix} x = \begin{bmatrix}6\\0\\6\end{bmatrix}$$

$$\begin{array}{r} \cancel{3x} + 3y - 3z = 6 \\ -\cancel{3x} - 7y + 11z = 6 \\ \hline -4y + 8z = 12 \end{array} \qquad \begin{array}{r} \cancel{3x} + 5y - 7z = 0 \\ \cancel{3x} \underset{(-)}{+} 3y \underset{(+)}{-} 3z = 6 \\ \hline 2y - 4z = -6 \\ -4y + 8z = 12 \\ \hline \end{array}$$

$$\begin{array}{r} -4y + 8z = 12 \\ 4y - 8z = -12 \\ \hline 0 \quad 0 \quad 0 \end{array}$$

Echelon form

$$\begin{bmatrix}3&3&-3&:6\\3&5&-7&:0\\-3&-7&11&:\end{bmatrix} \quad \begin{array}{l} R_2 \to R_2 - R_1 \\ R_3 \to R_3 + R_1 \end{array}$$

$$\begin{bmatrix}3&3&-3&:6\\0&2&-4&:-6\\0&-4&8&:12\end{bmatrix} x = \begin{bmatrix}6\\0\\6\end{bmatrix}$$

$-7+3$

$11-3$

$$\begin{bmatrix}3&3&-3&:6\\3&5&-7&:0\\3&-7&11&:6\end{bmatrix} \quad \begin{array}{l} \frac{R_1}{3} \\ R_2 - R_2 - 3R_1 \\ R_3 \to R_3 - 3R_1 \end{array} \Rightarrow \begin{bmatrix}1&1&-1&:2\\0&2&-4&:-6\\0&-4&8&:12\end{bmatrix}$$

$R_3 - R_3 - 2R_2$

$$= \begin{bmatrix}1&1&-1&:2\\0&2&-4&:-6\\0&0&0&:0\end{bmatrix}$$

$x \quad y \quad z$ → free variable

$x + y - z = 2$

$2y - 4z = -6$

$2y = -6 + 4z$

$\boxed{y = -3 + 2z}$

$x + (-3 + 2z) - z = 2$

$\boxed{x = -z + 5}$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} z + \begin{bmatrix} 5 \\ -3 \\ 0 \end{bmatrix}$$

Page No.
Date

12/25

*. Least Squares method :-

what if
→ $Ax = b$ have no solution

$$A^T A x = A^T b$$

→ Consider a matrix $A_{m \times n}$ and a non homogenous system $Ax \doteq b$ the least squares method is used to guess approx. solution x for the whole system when the system is inconsistent. This can be done by solving $A^T A x = A^T b$

1. Find the least squares solution of the system $Ax = b$ where $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix}$, & $b = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$ and also compute min. distance from the obtained solⁿ $\hat{X}$.

→ Consider $(A^T A) x = A^T b$

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}_{2\times3} \cdot \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix}_{3\times2} \cdot X = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}_{2\times3} \begin{bmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}_{3\times1}$$

$$\begin{bmatrix} 5 & 3 \\ 3 & 3 \end{bmatrix}_{2\times2} \cdot X = \begin{bmatrix} 0 \\ 6 \end{bmatrix}_{2\times1}$$

$$\begin{bmatrix} 5 & 3 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{pmatrix} 0 \\ 6 \end{pmatrix}$$

$$5x + \cancel{3y} = 0$$
$$-\ 3x + \cancel{3y} = 6$$
$$2x = -6$$
$$x = -3$$

$$5x + 3y = 0$$
$$15 + 3y = 0$$
$$y = 5$$

Page No.
Date

$\hat{x} = \begin{pmatrix} -3 \\ 5 \end{pmatrix}$

minimum distance

Given

$Ax = b$

Approximate sol $\hat{x}$

$A\hat{x} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix}_{3\times2} \begin{bmatrix} -3 \\ 5 \end{bmatrix}_{2\times1}$

$A\hat{x} = \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}_{3\times1}$

$(5, 2, -1) - (6, 0, 0)$

$d = (-1, 2, -1)$

$|d| = \sqrt{(-1)^2 + 2^2 + (-1)^2}$

$= \sqrt{6}$

2. Obtain an approx. solution by least squares method for the system $AX = b$ where $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & 2 & -3 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$

→ Consider $(A^T A)X = A^T b$

$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 1 & -1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & 2 & -3 \end{bmatrix} \cdot X = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 1 & -1 & -3 \end{bmatrix} \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$

$= \begin{bmatrix} 3 & 3 & -3 \\ 3 & 5 & -7 \\ -3 & 7 & 11 \end{bmatrix} X = \begin{bmatrix} 6 \\ 0 \\ 6 \end{bmatrix}$

Page No

Date

$$\begin{bmatrix} 3 & 3 & -3 \\ 3 & 5 & -7 \\ -3 & -7 & 11 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 6 \end{bmatrix}$$

Reduce to Eclaun Form

$3x + 3y - 3z = 6$

$3x + 5y - 7z = 0$

$-3x + 7y + 11z = 6$

$$\begin{bmatrix} 3 & 3 & -3 \\ 3 & 5 & -7 \\ -3 & -7 & 11 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 6 \end{bmatrix}$$

$R_2 \rightarrow R_2 - R_1$

$R_3 \rightarrow R_3 + R_1$

$$= \begin{bmatrix} 3 & 3 & -3 \\ 0 & 2 & -4 \\ 0 & -4 & 8 \end{bmatrix}$$

$R_3 \rightarrow R_3 + 2R_2$

$$= \begin{bmatrix} 3 & 3 & -3 \\ 0 & 2 & -4 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 6 \end{bmatrix}$$

$= [A : b]$

$$= \left[\begin{array}{ccc:c} 3 & 3 & -3 & 6 \\ 3 & 5 & -7 & 0 \\ -3 & -7 & 11 & 6 \end{array}\right] \rightarrow \begin{matrix} R_2 = R_2 - R_1 \\ R_3 = R_3 + 3R_1 \end{matrix}$$

$$= \left[\begin{array}{ccc:c} 3 & 3 & -3 & 6 \\ 0 & 2 & -4 & -6 \\ 0 & -4 & 8 & 12 \end{array}\right] \rightarrow R_3 \rightarrow R_3 + 2R_2$$

$$= \left[\begin{array}{ccc:c} 3 & 3 & -3 & 6 \\ 0 & 2 & -4 & -6 \\ 0 & 0 & 0 & 12 \end{array}\right] \rightarrow \frac{R_1}{3}$$

$$= \begin{array}{c} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 2 & -4 & -6 \\ 0 & 0 & 0 & 0 \end{array}\right] \\ \begin{array}{ccc} \uparrow & \uparrow & \uparrow \\ x & y & z \end{array} \end{array}$$

$z \rightarrow$ free variable

$\therefore$ eqn.

$$x + y - z = 2$$
$$2y - 4z = -6$$

$$2y - 4z = -6$$
$$2y = -6 + 4z$$
$$y = \frac{-6 + 4z}{2} = -3 + 2z$$

$$\cancel{2y}\ 2(-3+2z) - 4z = -6$$
$$-6 + 4z - 4z = -6$$
$$z = 0$$
$$x + (-3+2z) - z = 2$$
$$x - 3 + 2z - z = 2$$
$$x - z = 5$$
$$x = 5 + z$$

2. Use least square method for the following datapoint(s) find the approx. solution for the least
(0,6), (1,0) (2,0)

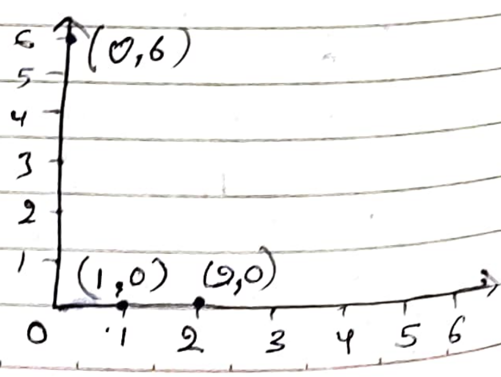
→ let $y = ax + b$ —— (1)

write given points

$(0,6) \Rightarrow 6 = a.0 + b$

$(1,0) \Rightarrow 0 = a.1 + b$

$(2,0) \Rightarrow 0 = a.2 + b$



Page No
Date

write in matrix form.

$$\underbrace{\begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}}_{B} = \underbrace{\begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix}}_{A} \cdot \underbrace{\begin{bmatrix} a \\ b \end{bmatrix}}_{X}$$

$B = AX$

i.e $AX = B$ form.

from $(A^T A)X = A^T B$.

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$$

Solve above eq^n as per ~~sus~~ example 1. we get

$$\hat{x} = \begin{pmatrix} -3 \\ 5 \end{pmatrix}$$

4.Q Consider a curve $y = \beta_0 + \beta_1 x$ which makes it good model for depend^n variable. Find the least square variable with the following data points.

(1, 1.8), (2, 2.7), (3, 3.4), (4, 3.8), (5, 3.9)

→ let $y = \beta_0 + \beta_1 x$

$1.8 = \beta_0 + \beta_1 1$

$2.7 = \beta_0 + \beta_1 2$

$3.4 = \beta_0 + \beta_1 3$

$3.8 = \beta_0 + \beta_1 4$

$3.9 = \beta_0 + \beta_1 5$.

write in matrix form.

$$\underbrace{\begin{bmatrix} 1.8 \\ 2.7 \\ 3.4 \\ 3.8 \\ 3.9 \end{bmatrix}_{5\times1}}_{B} = \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{bmatrix}_{5\times2}}_{A} \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}_{2\times1}}_{X}$$

Page No
Date

$$AX = B$$

Solve

$$(A^T A)X = A^T B$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{bmatrix} \cdot X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix} B$$

$$\begin{bmatrix} 5 & 15 \\ 15 & 1+4+9+16+25 \end{bmatrix} \cdot X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix} B$$

1
79
15
55

$$\begin{bmatrix} 5 & 15 \\ 15 & 55 \end{bmatrix} \cdot X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix} B \begin{bmatrix} 1.8 \\ 2.7 \\ 3.4 \\ 3.8 \\ 3.9 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 15 \\ 15 & 55 \end{bmatrix} \cdot X = \begin{bmatrix} 15.6 \\ 52.1 \end{bmatrix}$$

	3
1.8	1.8
5.4	2.7
10.2	3.4
15.2	3.8
19.5	3.9
	6

$$\begin{bmatrix} 5 & 15 \\ 15 & 55 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 15.6 \\ 52.1 \end{bmatrix}$$

$$5\beta_0 + 15\beta_1 = 15.6$$

$$5\beta_0 = 15.6 - 15\beta_1$$

$$\beta_0 = \frac{15.6 - 15\beta_1}{5}$$

$$\cancel{6}\ 15\beta_0 + 55\beta_1 = 52.1$$

$$\cancel{15\left(\frac{15.6 - 15\beta_1}{5}\right) + 55\beta_1 = 52.1}$$

$$5\beta_0 + 15\beta_1 = 15.6$$

$$-\ 15\beta_0 + 55\beta_1 = 52.1$$

$\beta_0 = 1.53$ $\beta_1 = 0.53$